

Normalisation

1. What is Normalisation?

Normalisation is the process of restricting database tables to minimize redundancies and reduce data anomalies, to bring databases to a consistent state. It is used as part of the design process or to modify existing data structures.

It works through a series of stages called normal forms.

2. When is a table 1NF?

A table is **1NF** when each column contains atomic (single values) with no repeating groups or multivalued attributes, making each row unique.

Example:

ID	Name	Phone
101	Alice	0467 406 293

3. When is a table 2NF?

A table is **2NF** if it is already in 1NF, but every non-key attribute is fully dependent on the entire primary key.

This mainly applies when the table had a composite primary key.

A table violates 2NF if a non-key attribute depends on only part of the composite key

Example:

Order Details: {Order ID, Product ID} (Key), Quantity

Products: Product ID (Key), Product Name

4. When is a table 3NF?

A table is **3NF** if it is already in 2NF and there are no transitive dependencies

A transitive dependency occurs when a non-key attribute depends on another non-key attribute.

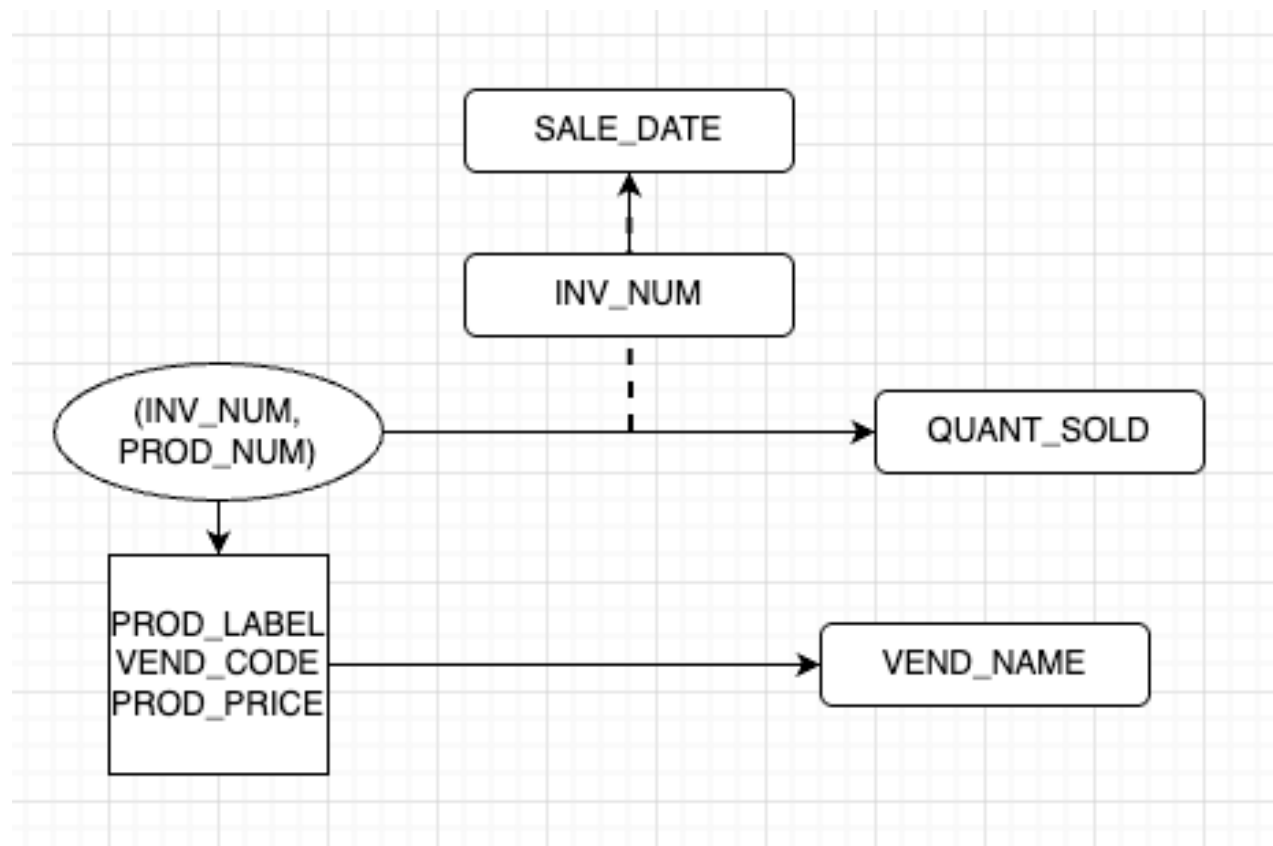
Example:

Table A: Customer (Customer ID [PK], Name, Postcode [FK])

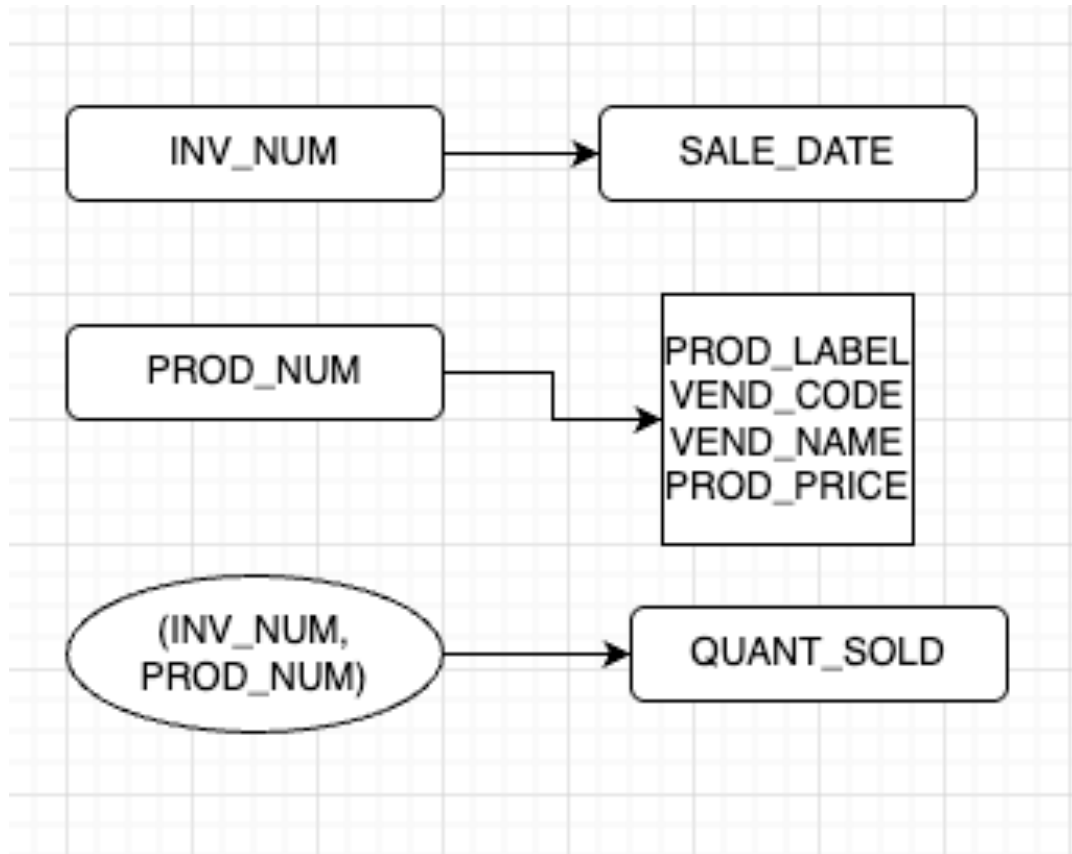
Table B: Postcode (Postcode [PK], City)

5. Draw a dependency table based on this table:

INV_NUM	PROD_NUM	SALE_DATE	PROD_LABEL	VEND_CODE	VEND_NAME	QUANT_SOLD	PROD_PRICE
211347	AA-E3522QW	15-Jan-2018	Rotary sander	211	NeverFail, Inc.	1	\$34.46
211347	QD-300932X	15-Jan-2018	0.25-in. Drill bit	211	NeverFail, Inc.	8	\$2.73
211347	RU-995748G	15-Jan-2018	Band saw	309	BeGood, Inc.	1	\$31.59
211348	AA-E3522QW	15-Jan-2018	Rotary sander	211	NeverFail, Inc.	2	\$34.46
211349	GH-778345P	16-Jan-2018	Power drill	157	ToughGo, Inc.	1	\$69.32



6. Remove all partial dependencies and draw the new dependency diagram



7. Remove all transitive dependencies and draw the new dependency diagram

